

Problem Statement ID	:	26024
Problem Statement	:	AI-Based Smart Governance and Compliance Monitoring System for Coal Mines
Title	:	MineKawach
Theme	:	Smart Automation
PS Category	:	Software
Team ID	:	138658
Team Name	:	Tesseract Titans

MINEKAWACH : AI-BASED SMART GOVERNANCE AND COMPLIANCE MONITORING SYSTEM FOR COAL MINES

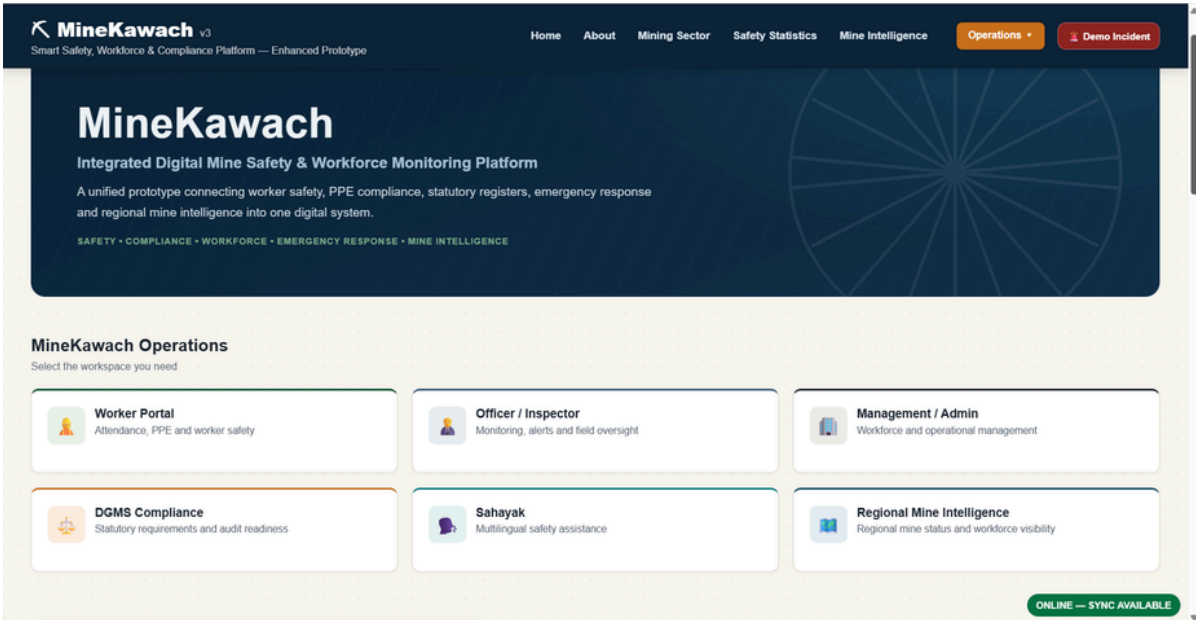


QR Code for MineKawach Prototype

PROTOTYPE DEMONSTRATION

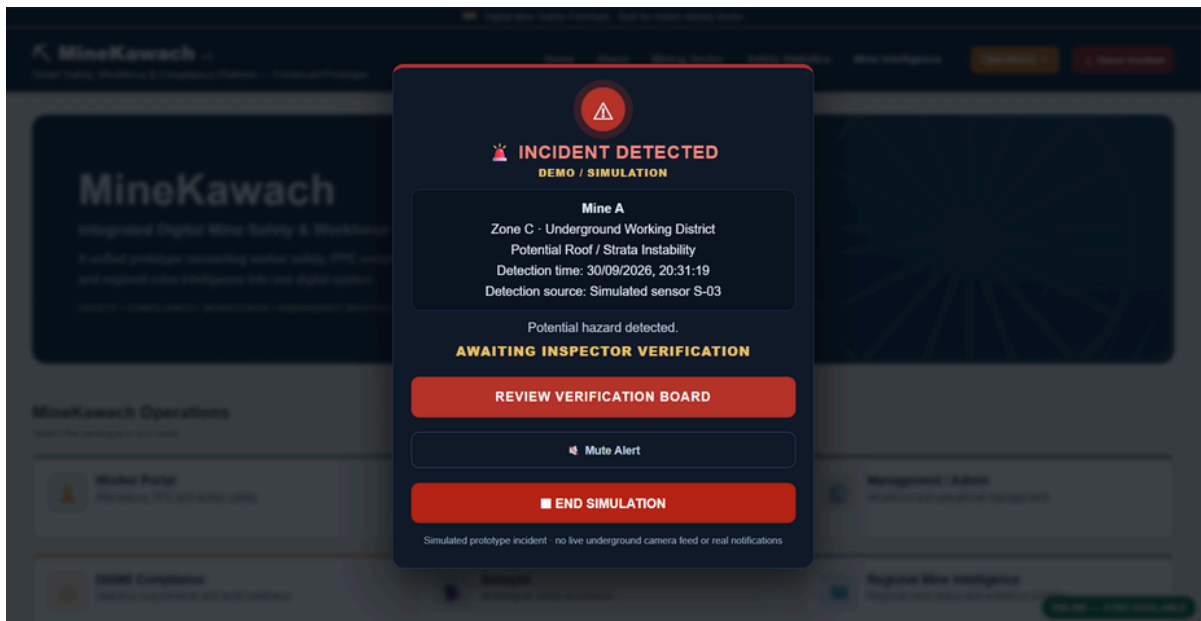
1. MineKawach Platform Overview

A unified digital platform connecting worker safety, workforce monitoring, compliance, emergency response and mine intelligence.



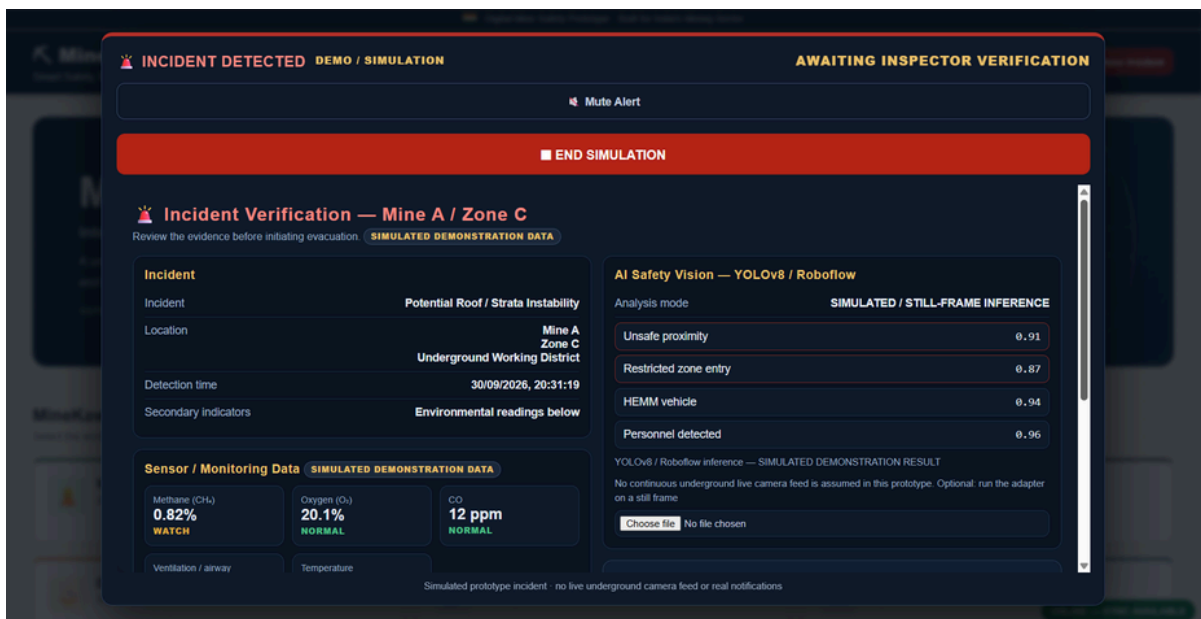
2. Incident Detection & Global Alert

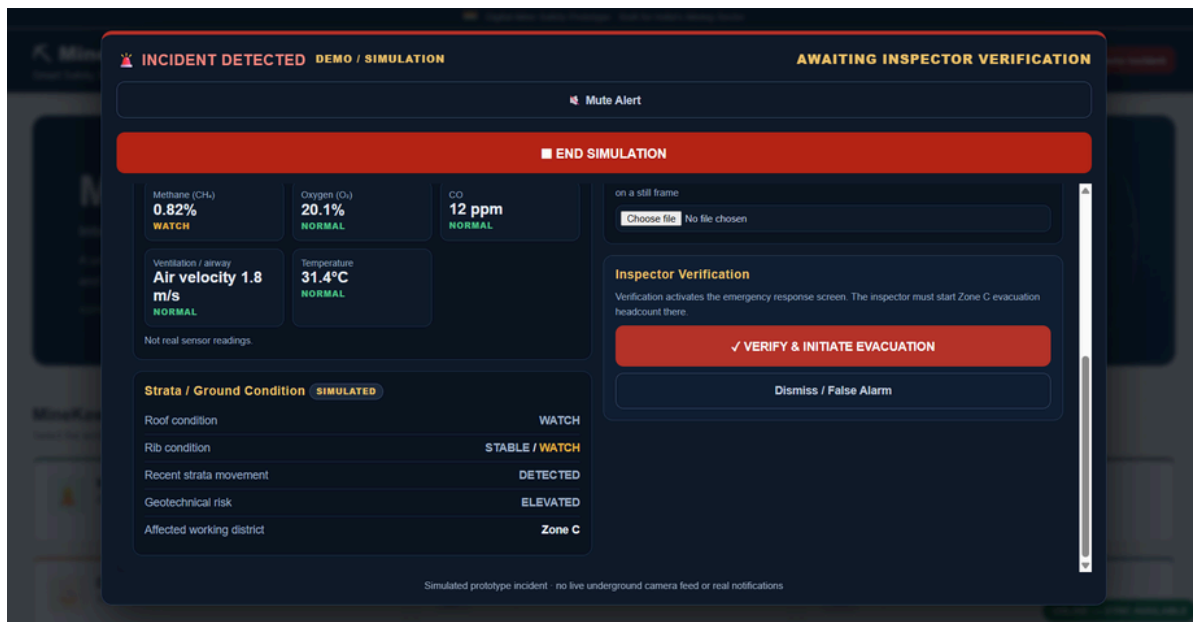
A simulated abnormal condition triggers a system-wide emergency state and routes the event for responsible-officer verification.



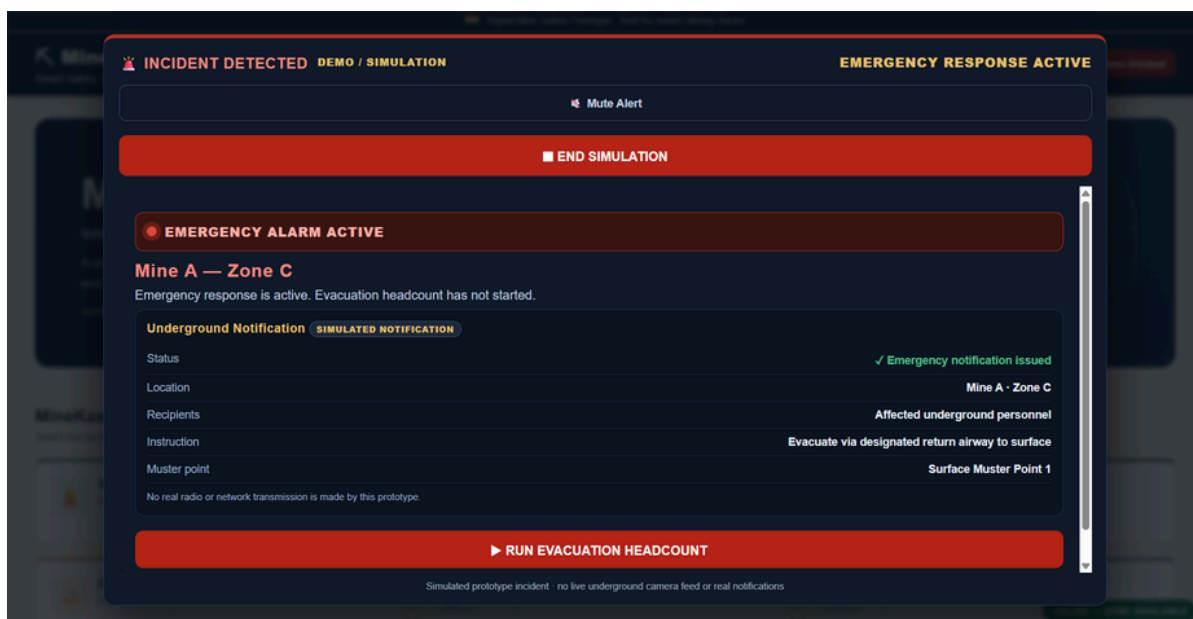
3. Human-in-the-Loop Incident Verification

Safety personnel review the detected event before initiating emergency evacuation procedures.



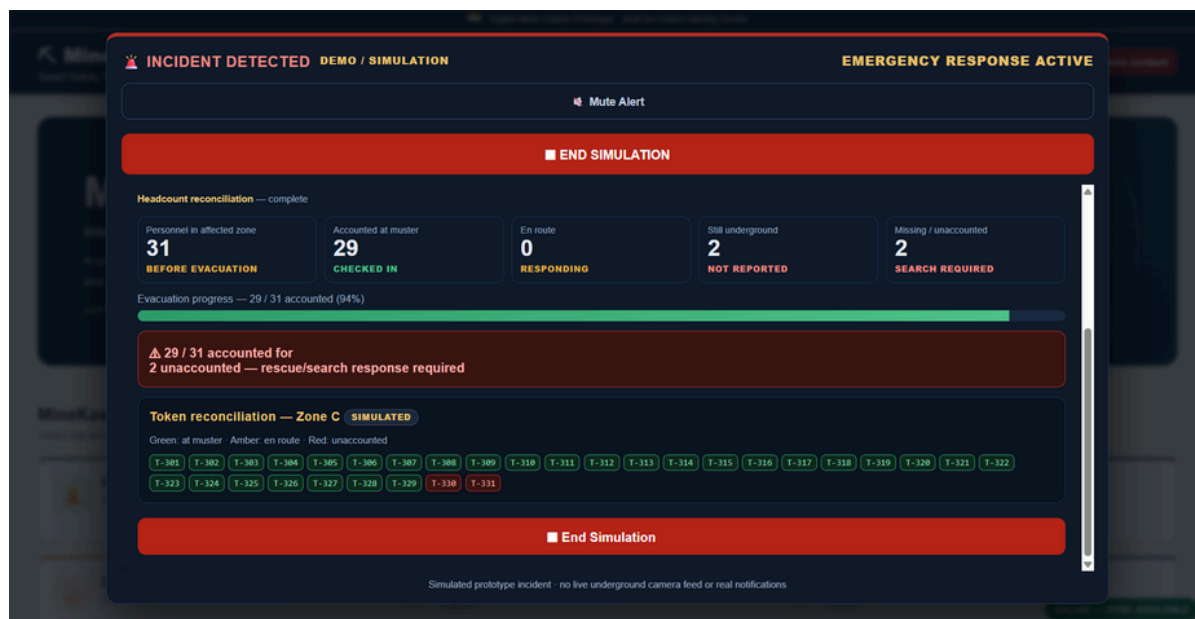
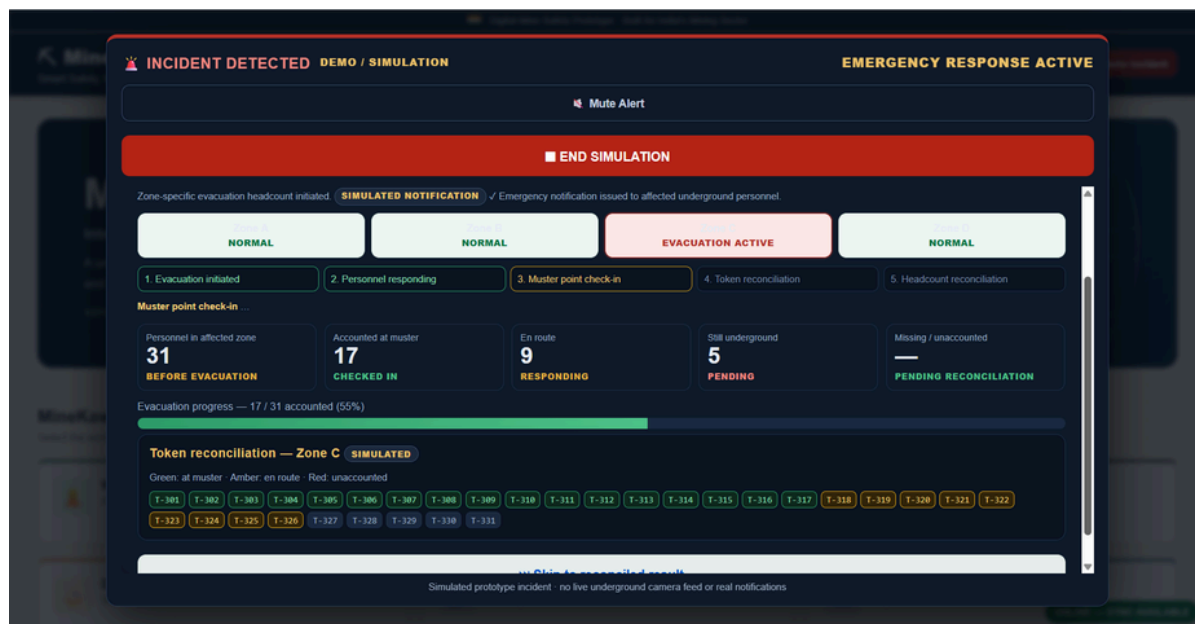


4. Emergency response is active.
Evacuation headcount has not started.



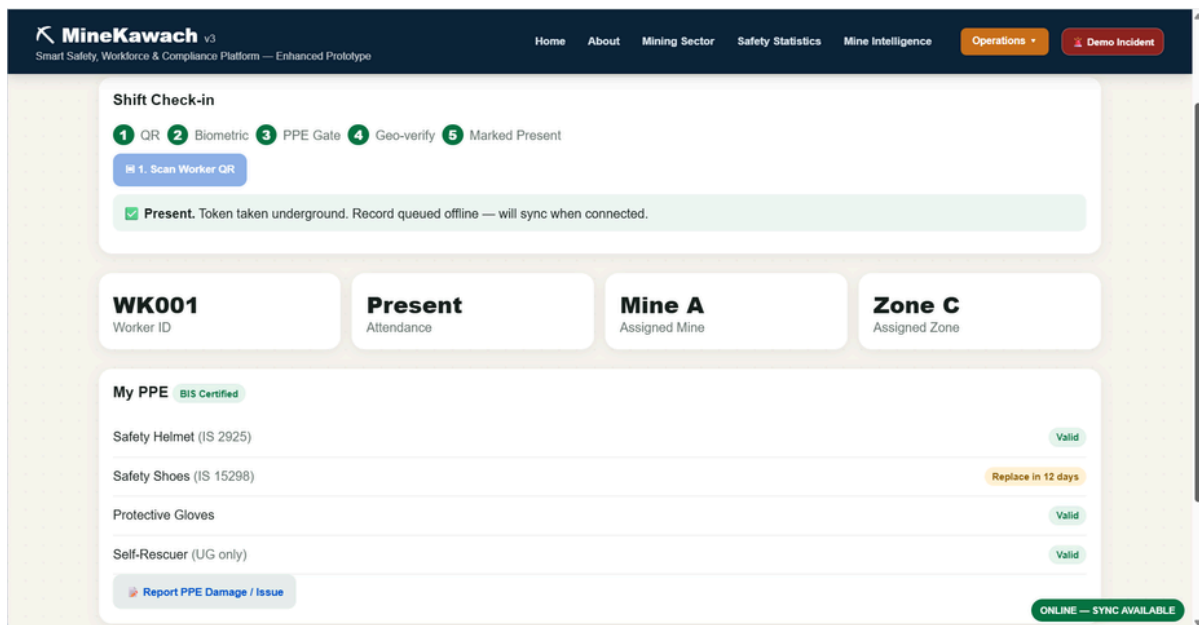
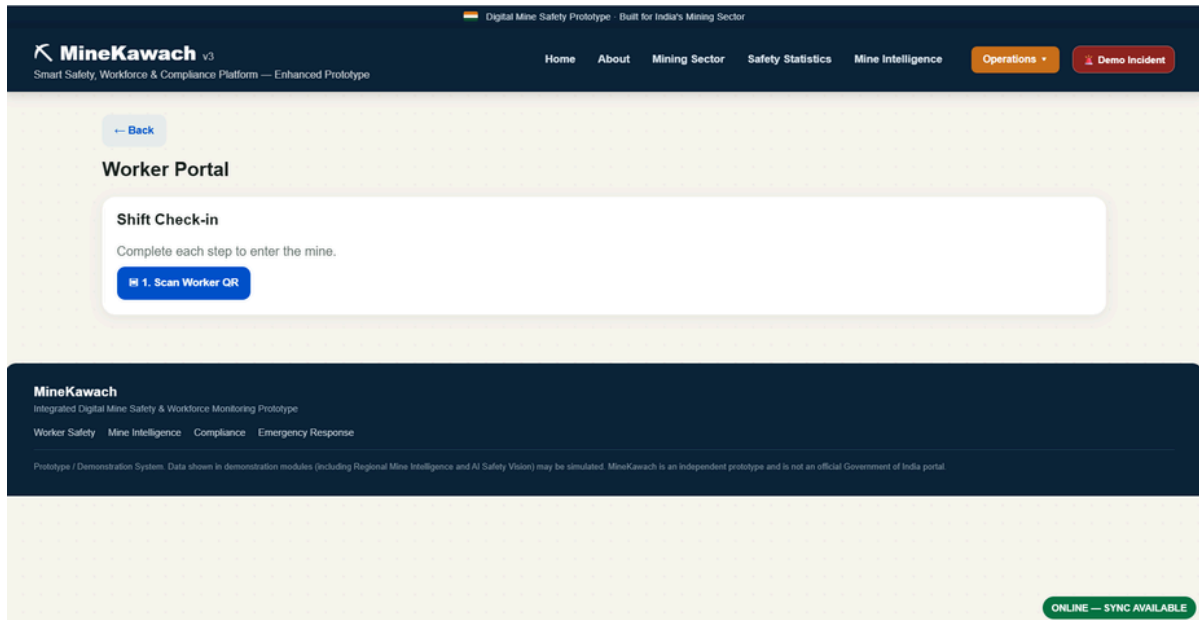
5. Zone-Specific Emergency Evacuation Headcount

Emergency muster begins only after incident verification and focuses on personnel in the affected zone.



6. Digital Worker Check-in & Workforce Identification

Enables structured worker entry, zone assignment and digital workforce presence tracking.



7. PPE Gate Verification

Digital verification of required personal protective equipment before mine entry, with PPE status linked to the worker record.

The screenshot shows the 'PPE Gate Verification' screen in the MineKawach v3 application. The header includes the MineKawach logo and navigation links: Home, About, Mining Sector, Safety Statistics, Mine Intelligence, Operations, and Demo Incident. The main content area is titled 'PPE Gate Verification' and displays a list of PPE items with their verification status. A summary section indicates that all PPE is verified and compliant with Rule 126. It also shows the location verification status and the last shift ended 14 hours ago. A 'Confirm & Mark Present' button is visible at the bottom left of the main content area. A green status bar at the bottom right indicates 'ONLINE — SYNC AVAILABLE'.

PPE Item	Status
Safety Helmet	✓ Verified
Safety Shoes	✓ Verified
Protective Gloves	✓ Verified
Self-Rescuer	✓ Verified
Reflective Jacket	✓ Verified

All PPE verified ✓ (Rule 126 compliant)
► Step 4: Geo-verifying assigned location...

Location verified: **Mine A / Zone C** matches assignment ✓
► Step 5: Fatigue check (HEMM operator)...

Last shift ended 14 hrs ago — fit for duty ✓
Confirm & Mark Present

ONLINE — SYNC AVAILABLE

8. Sahayak — Multilingual Safety Assistant

Provides workers with accessible safety assistance through multilingual voice

The screenshot shows the 'Sahayak — Multilingual AI Voice Assistant' screen in the MineKawach v3 application. The header includes the MineKawach logo and navigation links: Home, About, Mining Sector, Safety Statistics, Mine Intelligence, Operations, and Demo Incident. The main content area is titled 'Sahayak — Multilingual AI Voice Assistant' and features a purple icon of a person's head. Below the title, there is a description of the assistant's capabilities and a language selection dropdown menu. The language is currently set to 'English'. There are two buttons: 'Speak Greeting' and 'Ask by Voice'. Below these buttons, a green status bar indicates 'Assistant ready.' At the bottom, there is a section titled 'Quick questions' with four buttons: 'What PPE do I need today?', 'How many hours can I work?', 'How is my attendance linked to wages?', and 'What is the emergency helpline?'.

Sahayak — Multilingual AI Voice Assistant

Ask about safety rules, PPE, wages or attendance in your own language. Works offline-first; speech uses on-device engine where available.

Language / **English**

Speak Greeting **Ask by Voice**

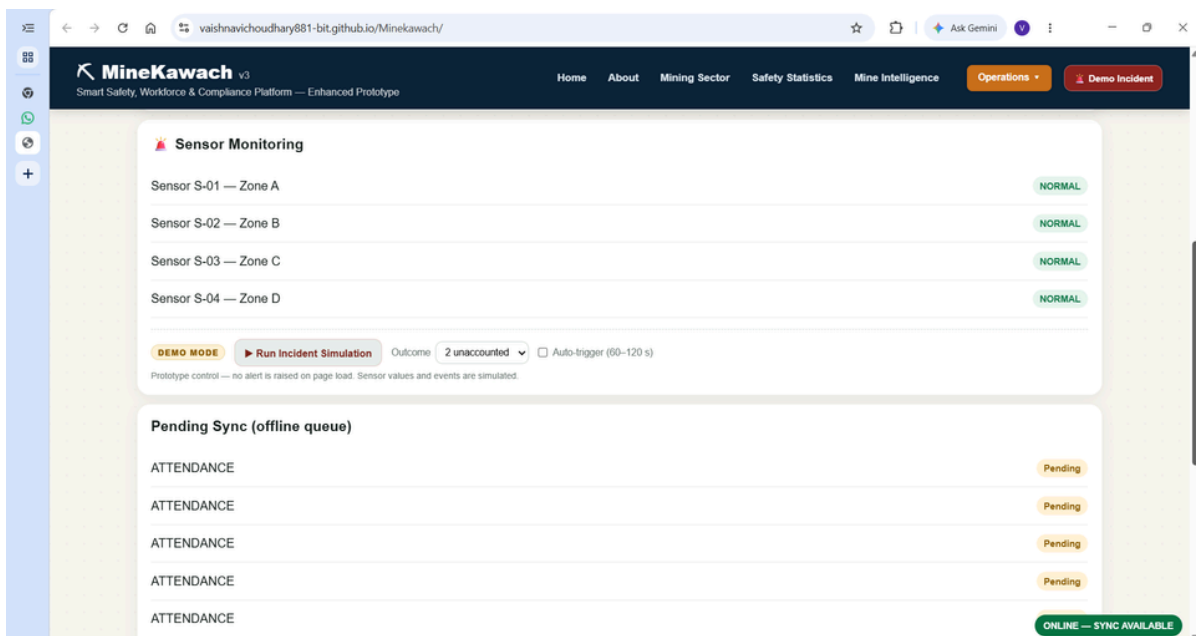
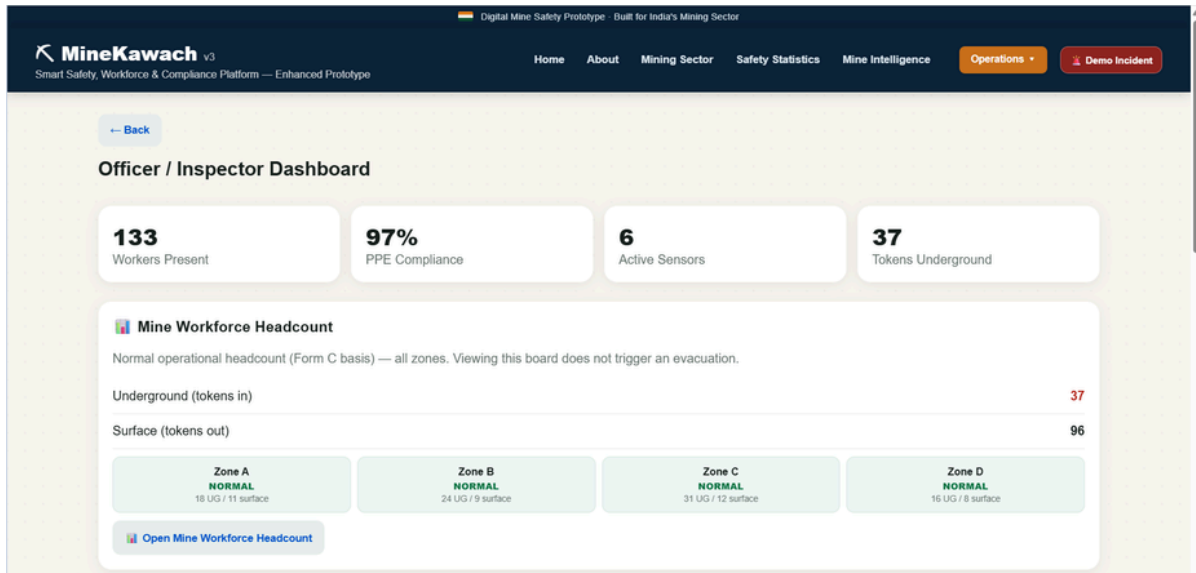
Assistant ready.

Quick questions

What PPE do I need today? **How many hours can I work?** **How is my attendance linked to wages?** **What is the emergency helpline?**

9. Officer / Inspector Dashboard

Provides a consolidated operational view of workforce presence, PPE compliance, underground personnel and monitored mine zones.



10. Normal Operational Headcount

Provides an all-zone view of workers underground and on the surface during routine operations. This remains separate from emergency evacuation headcount.

MineKawach v3
Smart Safety, Workforce & Compliance Platform — Enhanced Prototype

Home About Mining Sector Safety Statistics Mine Intelligence Operations Demo Incident

[-- Back](#)

Mine Workforce Headcount

Normal operational headcount — Mine A. Simulated prototype values. This is not the evacuation headcount.

ZONE	UNDERGROUND	SURFACE	TOTAL	STATUS
Zone A	18	11	29	Normal
Zone B	24	9	33	Normal
Zone C	31	12	43	Normal
Zone D	16	8	24	Normal
Total	89	40	129	

MineKawach
Integrated Digital Mine Safety & Workforce Monitoring Prototype
Worker Safety Mine Intelligence Compliance Emergency Response
Prototype / Demonstration System. Data shown in demonstration modules (including Regional Mine Intelligence and AI Safety Violator) may be simulated. MineKawach is an independent prototype and is not an official Government of India portal.

ONLINE — SYNC AVAILABLE

11. DGMS-Oriented Compliance Matrix

Maps MineKawach modules to applicable mining safety requirements, statutory records and compliance workflows.

MineKawach v3
Smart Safety, Workforce & Compliance Platform — Enhanced Prototype

Home About Mining Sector Safety Statistics Mine Intelligence Operations Demo Incident

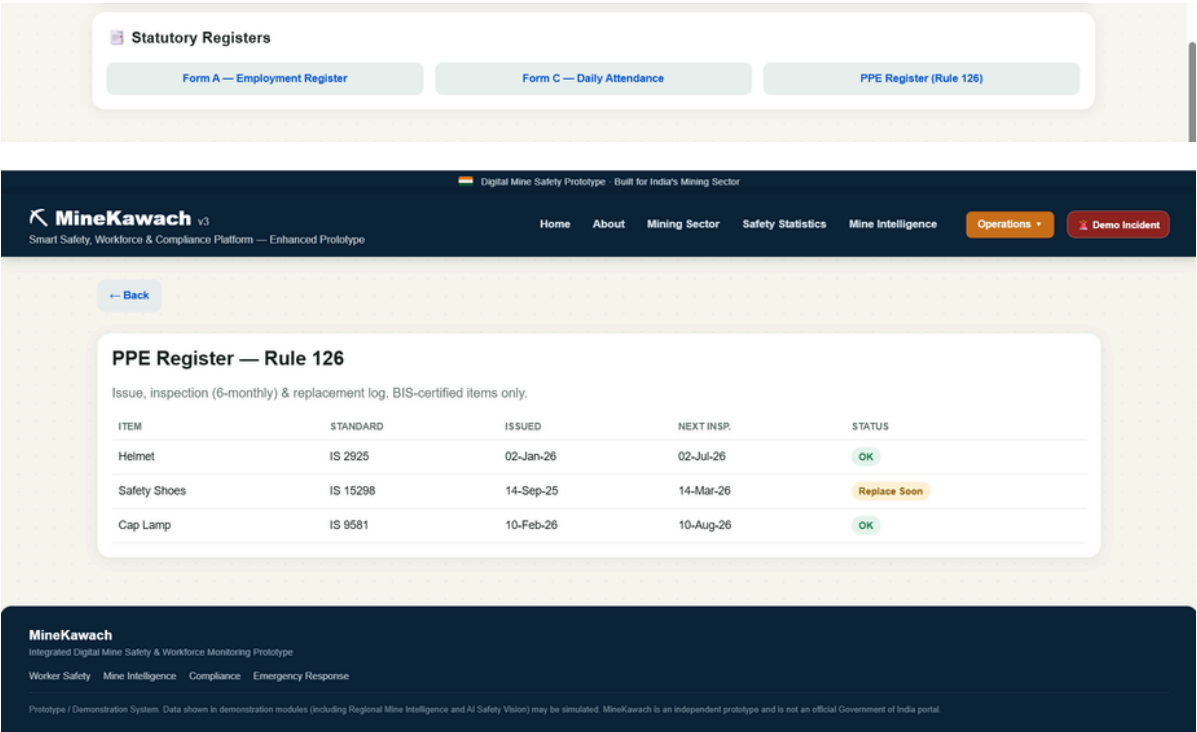
DGMS Compliance Matrix

Every MineKawach module mapped to the statute it satisfies — inspection-ready.

MODULE	REQUIREMENT	LAW / RULE	STATUS
Biometric + QR attendance	Biometric system; digitized Form-A; Aadhaar-based where possible	DGMS — 12th National Conference on Safety in Mines; Mines Rules 1955 (Forms A, C, D, E)	Covered
PPE gate verification + PPE register	Provide PPE, enforce use, 6-monthly inspection, replacement log	Rule 126, Mines Rules 1955; CMR 2017 Reg. 123	Covered
BIS-certified PPE tracking	Only BIS-certified PPE (e.g., IS 15298 footwear); test certificates retained	DGMS circular — PPE quality (BIS regime)	Covered
HEMM fatigue check-in/out	Max 8 hrs continuous duty, 30-min break after 4 hrs	CMR 2017; DGMS fatigue-monitoring directives	Covered
Token headcount board	Know exactly who is underground at any time; rescue headcount	Mines Rules 1955; CMR 2017 (manager's duties)	Covered
Skill-gate at check-in	No engagement without assessed skill/competency	Mines Act 1952; DGMS skill-assessment circulars	Covered
Contractor coverage	Contract workers in Form A, same PPE & benefits	Mines Act 1952; Mines Rules 1955	Covered
Alcohol prohibition enforcement	No alcohol/drugs below ground; checks at entry	Mines Rules 1955	Roadmap
Training records	Initial & refresher training, MVT Rules compliance	Mines Vocational Training Rules 1966	Roadmap
Medical fitness (IME/PME)	Initial & periodic medical examinations	Mines Rules 1955	Roadmap

ONLINE — SYNC AVAILABLE

MineKawach brings employment, attendance and PPE records into a structured digital workflow to improve traceability and inspection readiness.



Extends MineKawach from individual-worker safety to multi-mine operational visibility using demonstration data.

